HEEJO JEONG

wjdgmlwh1629343@gmail.com github homepage

RESEARCH INTERESTS

Computer Graphics, Rendering, Physics-based Animation, Numerical Methods, Optimization

EDUCATION

Korea University Seoul, Republic of Korea
M.S. in Computer Science and Engineering *Sep. 2023 - Aug. 2025*

- Advisor: Prof. JungHyun Han
- GPA: 4.34 / 4.5

Korea University Seoul, Republic of Korea
B.S. in Artificial Intelligence (Interdisciplinary Program) *Sep. 2019 - Aug. 2023*
B.S. in Civil, Environmental and Architectural Engineering *Mar. 2017 - Aug. 2023*

- Including two years of military service
- GPA: 4.06 / 4.5

PUBLICATIONS

[1] Momentum-preserving Inversion Alleviation for Elastic Material Simulation
Heejo Jeong, Seung-wook Kim, JaeHyun Lee, Kiwon Um, Min Hyung Kee, and JungHyun Han, Computer Animation and Virtual Worlds (CAVW), Vol. 35, No. 3, May 2024, pp. e2249.

- Poster version presented at the Korea Computer Graphics Society (KCGS), July, 2024.

RESEARCH & PROJECT

Learning Neural Hyper-elastic Constraints in XPBD Simulation
Researcher *Jul. 2024 – Feb. 2025*

- Implemented a differentiable XPBD solver for learning hyper-elastic constraints from a single motion trajectory.

Constraining Velocities in Position-based Dynamics
Research Assistant *Mar. 2024 – May 2024*

- Developed a velocity-based extension of the PBD framework to enhance stability and efficiency in dynamic simulations.
- Submitted to SIGGRAPH Asia 2024.

Dimension Expansion for Finite Element Method-based Deformable Body Simulation
Researcher *Sep. 2023 – Dec. 2023*

- Extended a finite element method-based deformable body simulation into 4D by introducing auxiliary coordinate and projecting the result back to 3D for rendering.

LG Electronics: Real-time Air Conditioning Airflow Simulation and Visualization
Project Assistant *Jun. 2023 – Sep. 2023*

- Implemented a Poisson solver using preconditioned conjugate gradient method with a sparse matrix for fluid simulation.

Real-time Vision-based Human Pose Matching Framework
Research Assistant *Mar. 2023 – May 2023*

- Implemented a multiple human tracking module using YOLOv7 and Kalman filter.

EXPERIENCE

Research Associate, Korea Univ., Korea *Sep. 2025 – Present*

Research Intern, Télécom Paris, Institut Polytechnique de Paris, France *Jan. 2025 – Feb. 2025*

Research Intern, Korea Univ., Korea *Sep. 2023 – Aug. 2024*

SCHOLARSHIPS

Research Assistant Scholarship, Korea University *Fall 2023 – Spring 2025*

Research Scholarship, Korea University *Fall 2023 – Fall 2024*

BK21 FOUR Outstanding New Student Scholarship *Fall 2023*

National Grant *Spring 2017 – Fall 2019, Fall 2022 – Spring 2023*

KU Alumni Scholarship, Korea University *Spring 2023*

Work Scholarship, Korea University *Fall 2022, Spring 2023*

POSCO Scholarship, POSCO *Fall 2022*

Special Scholarship, Korea University *Spring 2019*

Study Scholarship, Korea University *Fall 2018, Fall 2019*

HONORS

Semester High Honors *Spring 2018, Fall 2018, Fall 2022, Spring 2023*

TECHNICAL SKILLS

Languages: C/C++, Python, MATLAB

APIs: OpenGL, OpenMP, CUDA, Taichi Lang , PyTorch

Software: LaTeX, Blender, MS Office, Photoshop

LANGUAGE LEVEL

Korean: Native

English: Fluent (TOEFL iBT 97)